

GENETICS & MOLECULAR BIOLOGY REVISION NOTES

PU-II / NEET Quick-Recap Formula & Concept Sheets

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PART 1: PRINCIPLES OF INHERITANCE & VARIATION

1. Seven Pea Contrasting Characters

- **Height:** Tall (Dom) vs. Dwarf (Rec)
- **Flower Colour:** Violet (Dom) vs. White (Rec)
- **Flower Position:** Axial (Dom) vs. Terminal (Rec)
- **Pod Shape:** Inflated (Dom) vs. Constricted (Rec)
- **Pod Colour:** Green (Dom) vs. Yellow (Rec)
- **Seed Shape:** Round (Dom) vs. Wrinkled (Rec)
- **Seed Colour:** Yellow (Dom) vs. Green (Rec)

2. Non-Mendelian Extensions

- **Incomplete Dominance:** *Antirrhinum / Mirabilis* → **1:2:1** (Red:Pink:White).
- **Co-dominance:** Both alleles expressed → $I^A I^B$ (AB Blood), Cattle Roan coat.
- **Multiple Allelism:** ABO Blood (I^A , I^B , i) → **6 Genotypes, 4 Phenotypes**.
- **Pleiotropy:** 1 Gene → Many Traits (PKU, Starch size in pea).
- **Polygenic:** Many Genes → 1 Trait (Human skin colour: *AABBCC* to *aabbcc*).

3. Standard Genetic Ratios

Cross Type	Phenotypic	Genotypic
Monohybrid	3 : 1	1 : 2 : 1
Incomplete Dom.	1 : 2 : 1	1 : 2 : 1
Monohybrid Test	1 : 1	1 : 1
Dihybrid Cross	9 : 3 : 3 : 1	1:2:1:2:4:2:1:2:1
Dihybrid Test	1 : 1 : 1 : 1	1 : 1 : 1 : 1

4. Linkage & Sex Determination

- **Morgan's Drosophila:** y-w tight linkage (**1.3% rec**); w-m loose linkage (**37.2% rec**).
- **Alfred Sturtevant:** $1\% \text{ ext}\{ \text{Recombination} \} = 1 \text{ ext}\{ \text{cM (Map Unit)} \}$.
- **XX-XY:** Humans, *Drosophila* (Male heterogametic).
- **XX-XO:** Grasshopper / Insects (Male heterogametic).
- **ZW-ZZ:** Birds / Fowls (Female heterogametic: ZW ♀).
- **Haplodiploidy:** Honeybee (♀ = 32 diploid, ♂ = 16 haploid).